AI-Powered Analog Deep Learning Hardware Accelerator Design

Track: D: AI4AIchips

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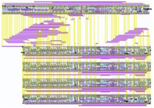

Acknowledgments

Dr. Francesco Stefanni, former labmate of Zhiyang Ong at University of Verona, who helped him craft LaTeX presentation slides.

The engineers of Mythic AI (Austin, TX) for showing us that analog in-memory computing is plausible, and their inspirational grit, from refusing to let their start-up go bankrupt multiple times, motivates us to be resilient in completing the IEEE SSCS Chipathon 2026 project.

Prof. Kaushik Sengupta at Princeton University and Prof. David Z Pan at The Awesome and Mighty University of Texas at Austin (trolling Aggieland haters, Go Longhorns!!!., Hook'em horns!!!) for inspiring us to craft AI-powered electronic design automation tools to kill IC design jobs.

Project Information

Goal and Application

- Goals:
  - To develop a multi-modal AI-powered analog or radio frequency (RF) IC design flow
  - To apply the aforementioned AI-powered IC design flow for designing analog deep learning hardware accelerators
- Application: Analog deep learning hardware accelerators can be used for environmental monitoring, by training on analog signals captured from the sensors with data conversion between analog and digital signals.

Project Information

Design – High Level Proposal (2-3 sentences + Image)

- Use semiconductor device models of memristors and floating-gate MOSFETs to approximate memristive and floating-gate MOSFET -based analog in-memory computing systems
- Implement analog in-memory computing systems by replacing SRAM cell in SRAM subsystem with approximate RAM cell circuit model, based on the aforementioned semiconductor device models
- Use recursive self-improvement to design the analog in-memory computing systems, where hardware generators are created to train the AI model (from memory subsystem with memristors or floating-gate MOSFETs to analog in-memory computing systems)

Project Information

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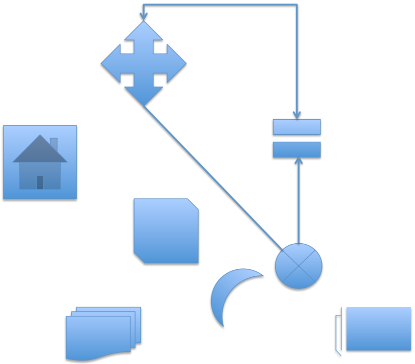

Figure: My caption [?, ?]

Slide Title 5

Slide Subtitle 5.

Statement 1 [?].

Statement 2.

Statement 3.

Team Members' Experiences

- Zhiyang Ong:
  - 32-kbit synchronous SRAM with 32-bit words
  - MIPS processor architectures: single cycle, multi-cycle, 5-stage pipelined, 128-bit
  - Viterbi decoder, and encoder/decoder pairs for linear code
  - 64-bit Ladner-Fischer tree adder, using skew tolerant domino logic and logical effort
  - FIR filter from the bilinear transformation of an analog Butterworth filter
  - high frequency Colpitts oscillator, with an operating frequency of 20 MHz
  - audio amplifier: power & pre- amplifiers, active filters, & RC-filter bridge rectifier
  - EDA projects: gate sizing,, SRAM CAD for design space exploration, hardware generators for encoder/decoder pairs targeting linear code, logic simulator, PODEM ATPG

Slide Title 6

Slide Subtitle 6.

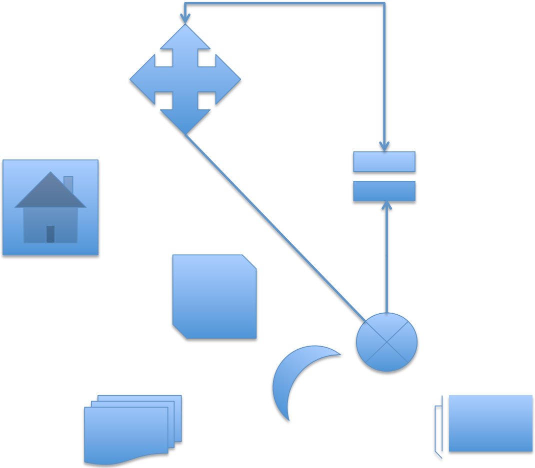

Figure: Caption #2

Questions, Suggestions, Doubts?

Statement 1 [**?**].

Statement 2.

Statement 3.

References

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